



Chinook Project: SHM Circuit Design & Grounding Strategy

KiCad Contest Technical Documentation

1. Circuit Architecture: The Wheatstone Bridge

To measure the tiny resistance changes in the carbon fiber lugs (R_LUG), the circuit uses a bridge configuration to convert resistance into a measurable voltage (V_OUT).

Components:

- **R1, R2, R3:** High-precision (0.1%) fixed resistors. These should match the nominal resistance of your carbon fiber joint (e.g., 100 Ohms).
- **R_LUG:** The active carbon fiber joint sensor.
- **Excitation Voltage (V_EXC):** A stable 3.3V or 5V source from the motor controller.

Formula: $V_{OUT} = V_{EXC} * [(R_{LUG} / (R3 + R_{LUG})) - (R2 / (R1 + R2))]$

2. Signal Conditioning: Instrumentation Amplifier

The raw V_OUT signal is extremely small. An **INA126** or **AD8226** Instrumentation Amplifier is used to:

- Provide high Input Impedance (so as not to 'load' the sensor).
- Amplify the signal to a 0-3.3V range suitable for an Analog-to-Digital Converter (ADC).

3. Integrated Grounding & ESD Mitigation

To ensure signal integrity and safety, the circuit implements a tiered grounding strategy:

- **Analog Ground (AGND):** Dedicated ground for the Wheatstone bridge to minimize noise.
- **Digital Ground (DGND):** Common ground for the MCU and motor drivers.
- **Static Drain Path:** A high-impedance path (1-10 Megaohm resistor) connecting the Carbon Fiber Lug mesh to the **Earth Ground Tail-Strap**. This drains static without creating a low-resistance path for accidental battery shorts.

4. Innovation Note for Judges

“By utilizing the inherent conductivity of the structural carbon fiber as a sensing element, the Chinook eliminates the weight and complexity of discrete strain gauges. The integration of a high-impedance static drain further ensures that the structural dashboard remains resilient to environmental electrical noise.”